

Commonly Used Distributions

Lecture 9

Topics in this lecture

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- Commonly used discrete distributions
- Commonly used continuous distributions

Discrete Distributions

- Bernoulli
- Binomial
- Geometric
- Poisson

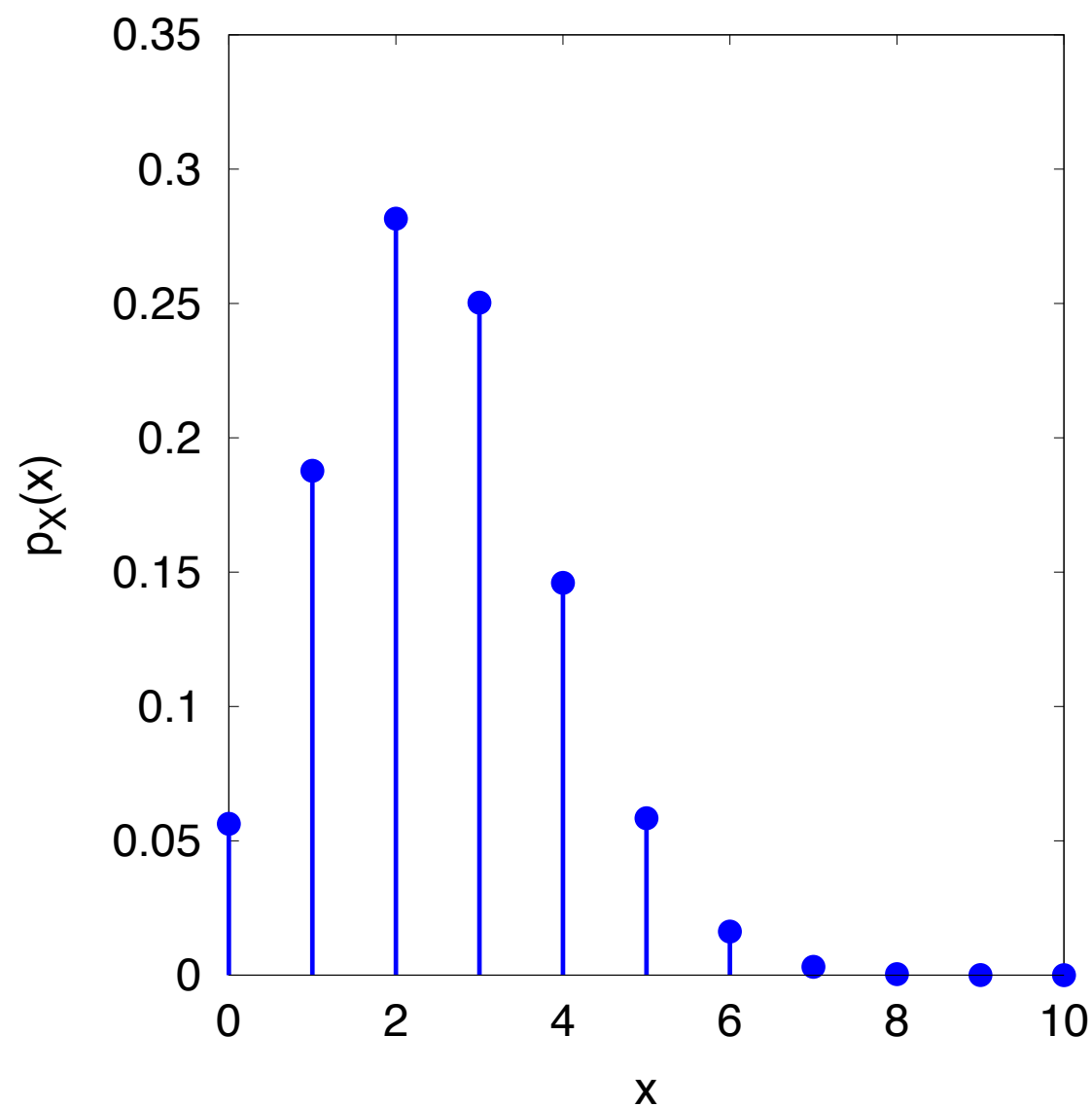
Bernoulli Distribution

Notation	$X \sim \text{Bernoulli}(p)$ <i>↗ Probability</i>
PMF	$p_X(x) = \begin{cases} p & , x = 1 \\ 1 - p & , x = 0 \\ 0 & , \text{otherwise} \end{cases}$
Mean	$\mu_X = p$
Variance	$\sigma_X^2 = p(1 - p)$

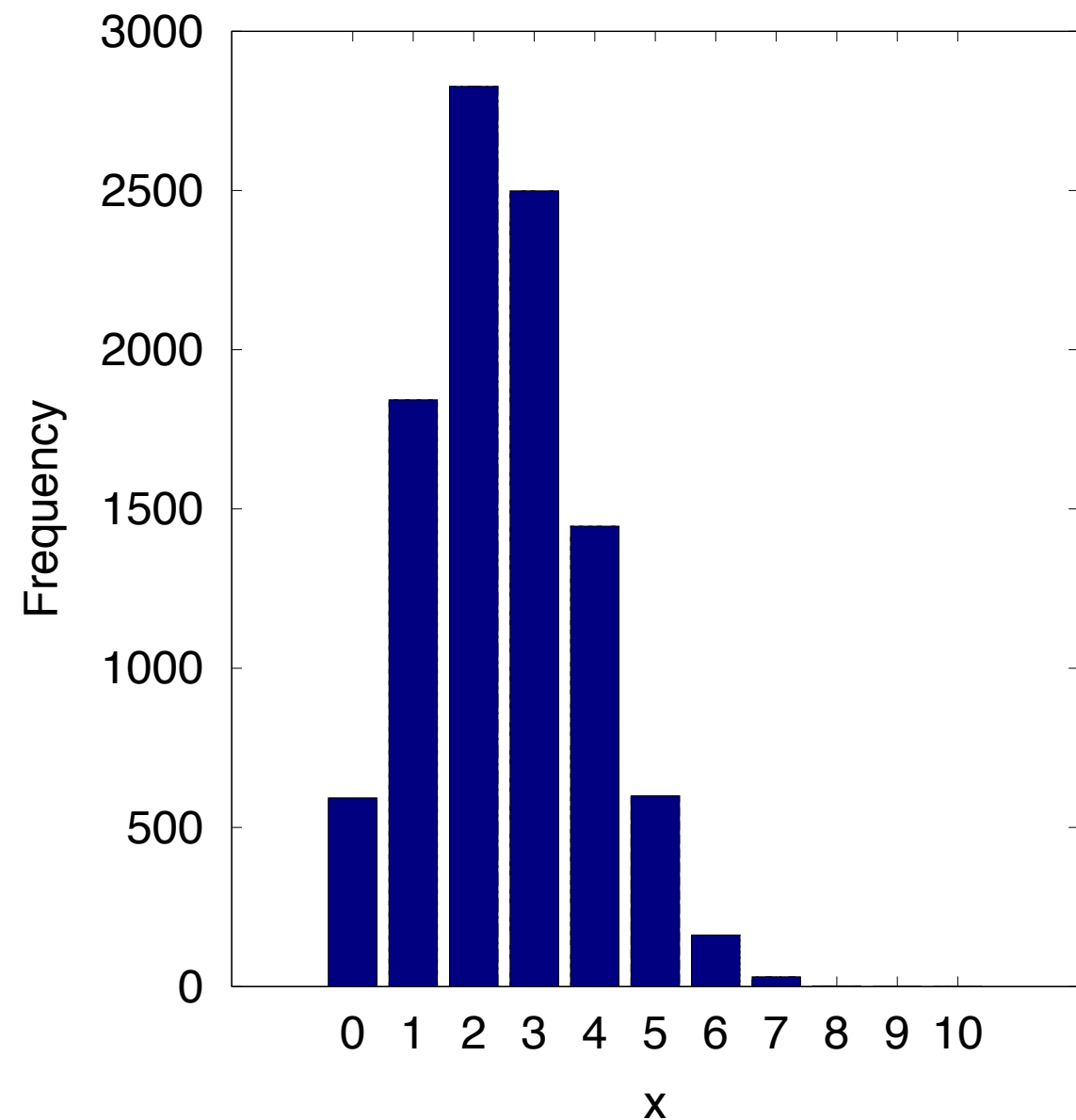
Binomial Distribution

Notation	$X \sim \text{Binomial}(n, p)$ <i>Handwritten notes:</i> จ.บ. A มี 9 ที่ทดลอง $\swarrow \searrow$ A. หนึ่งครั้งหนึ่งที B. ทดลอง 9 ครั้ง สักหนึ่ง
PMF	$p_X(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & , x \in \{0, 1, 2, \dots, n\} \\ 0 & , \text{otherwise} \end{cases}$
Mean	$\mu_X = np$
Variance	$\sigma_X^2 = np(1-p)$

Binomial Distribution



PMF of Binomial(10,0.25)

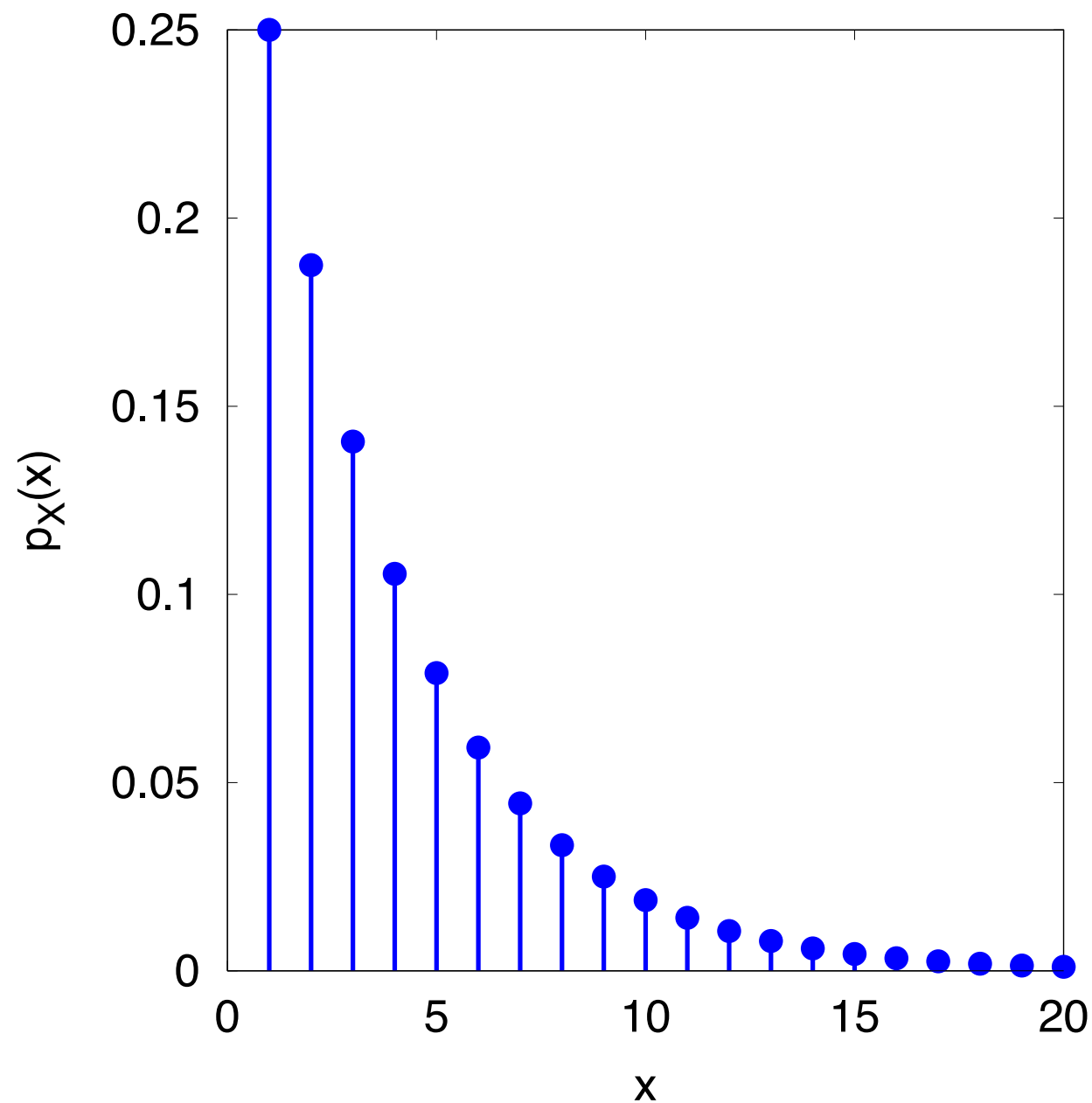


Histogram of samples drawn from Binomial(10,0.25)

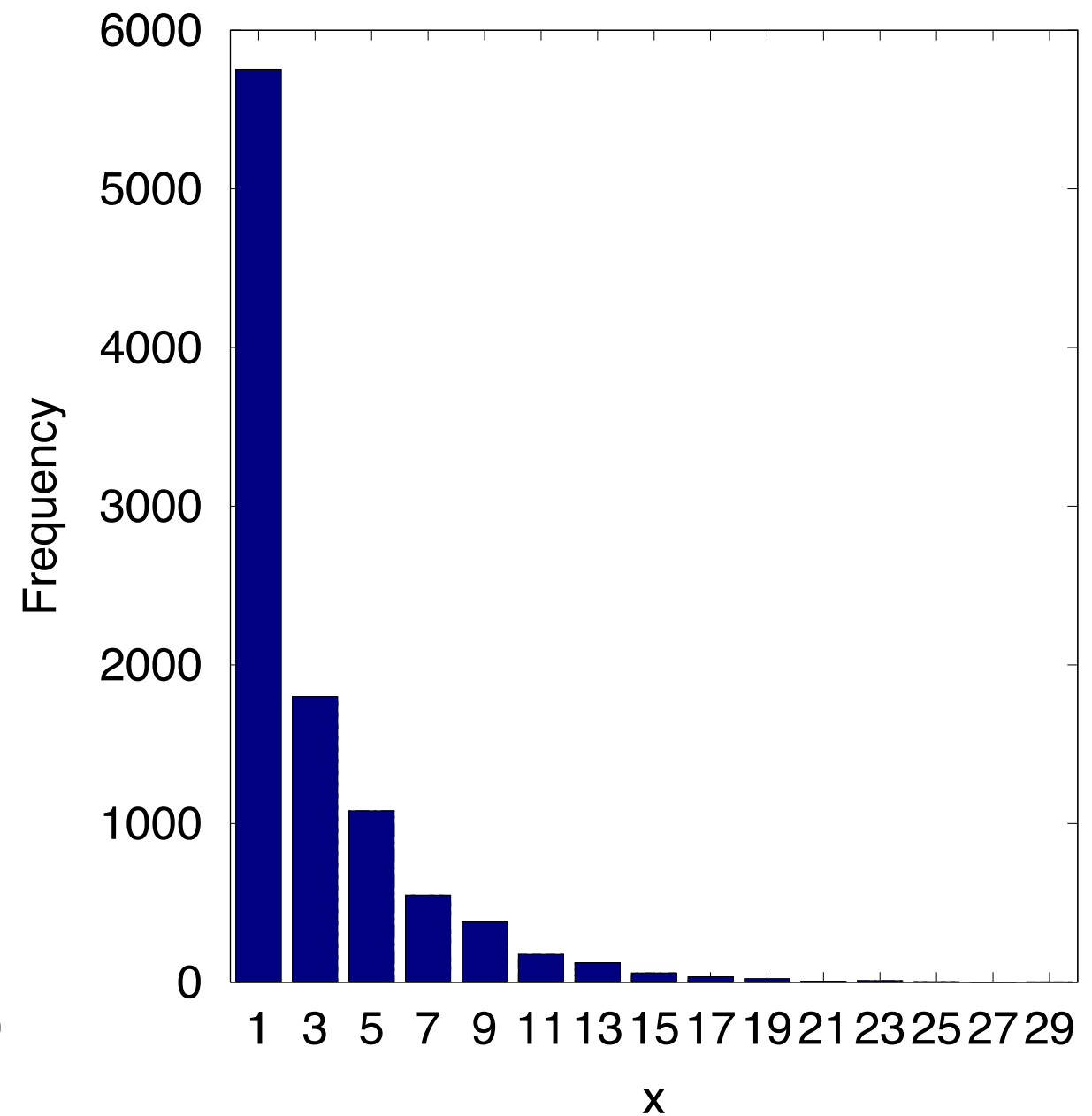
Geometric Distribution

Notation	$X \sim \text{Geometric}(p)$
PMF	$p_X(x) = \begin{cases} p(1-p)^{x-1} & , x \in \{1, 2, 3, \dots\} \\ 0 & , \text{otherwise} \end{cases}$
Mean	$\mu_X = \frac{1}{p}$
Variance	$\sigma_X^2 = \frac{1-p}{p^2}$

Geometric Distribution



PMF of Geometric(0.25)

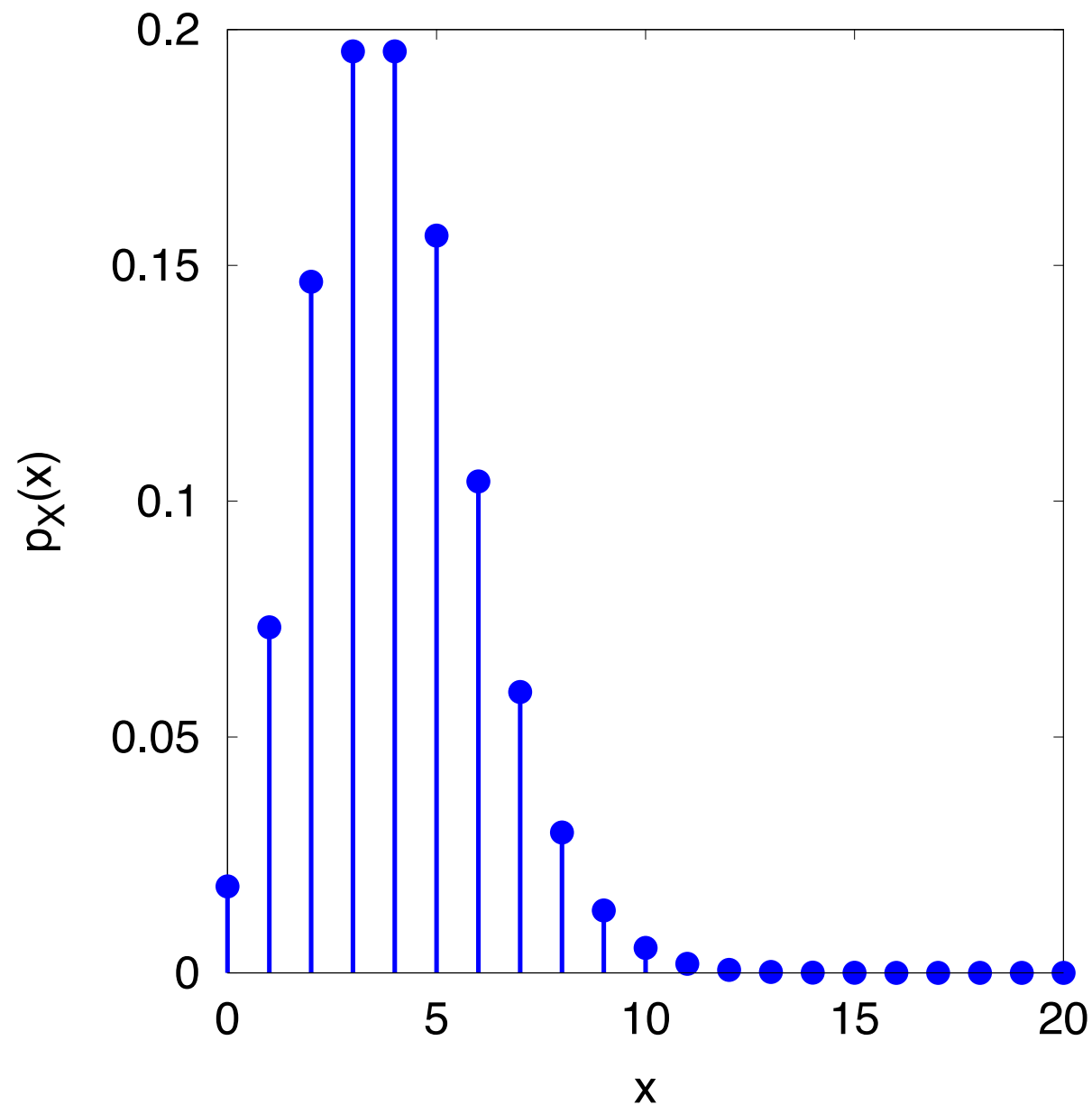


Histogram of samples drawn from Geometric(0.25)

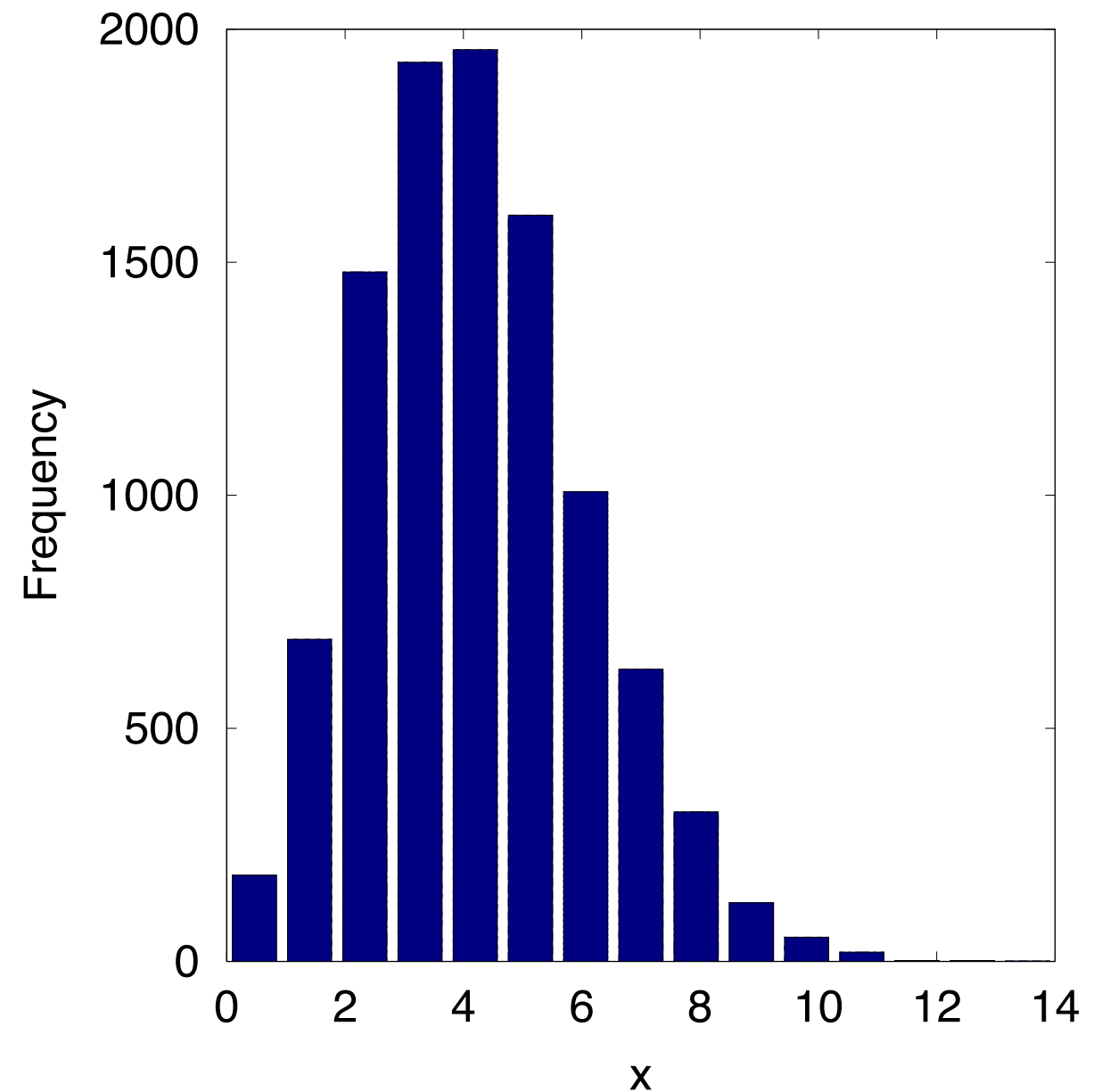
Poisson Distribution

Notation	$X \sim \text{Poisson}(\lambda)$
PMF	$p_X(x) = \begin{cases} \frac{\lambda^x}{x!} e^{-\lambda} & , x \in \{0, 1, 2, \dots\} \\ 0 & , \text{otherwise} \end{cases}$
Mean	$\mu_X = \lambda$
Variance	$\sigma_X^2 = \lambda$

Poisson Distribution



PMF of Poisson(4)



Histogram of samples drawn from Poisson(4)

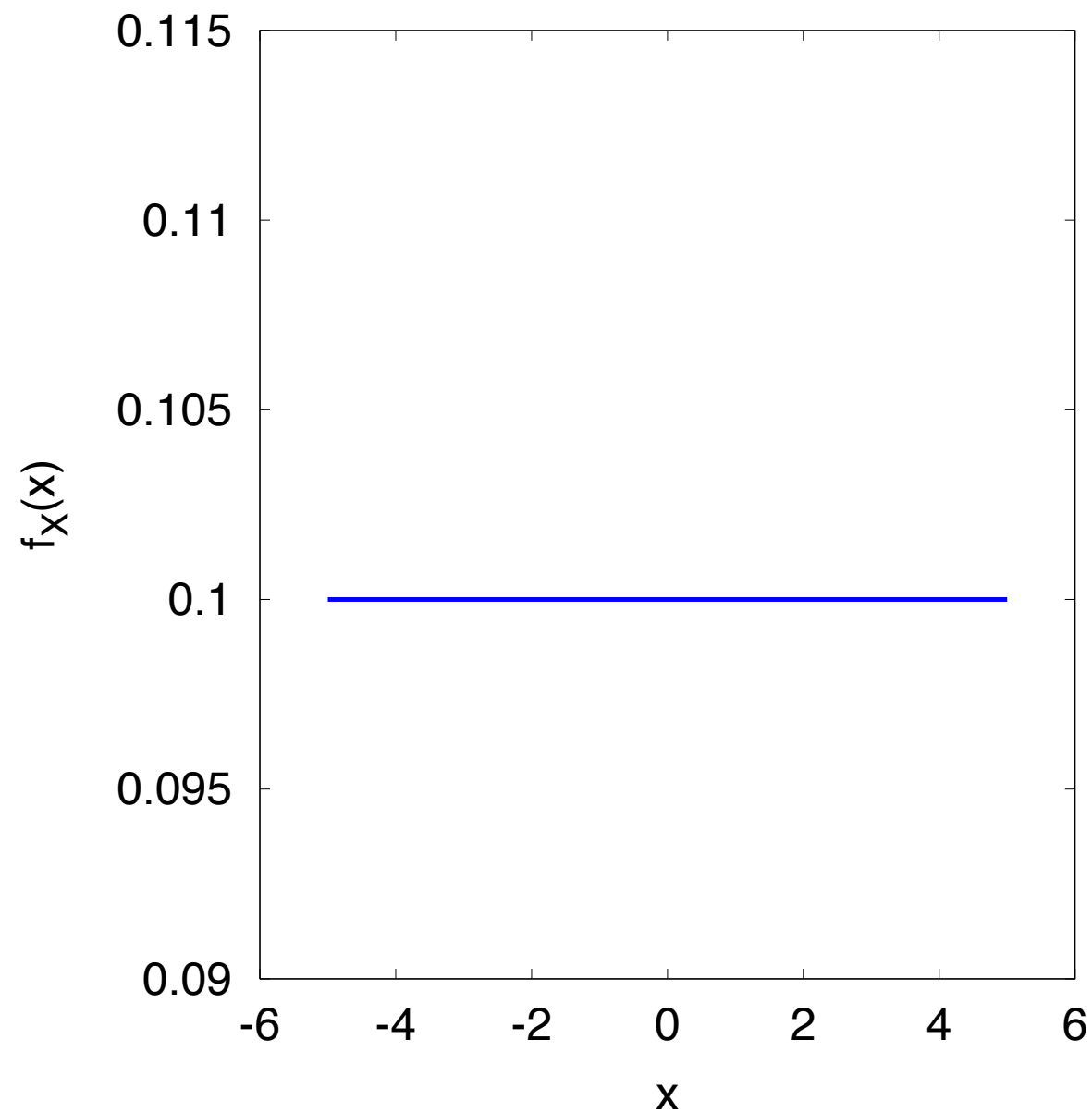
Continuous Distributions

- Uniform
- Exponential
- Normal

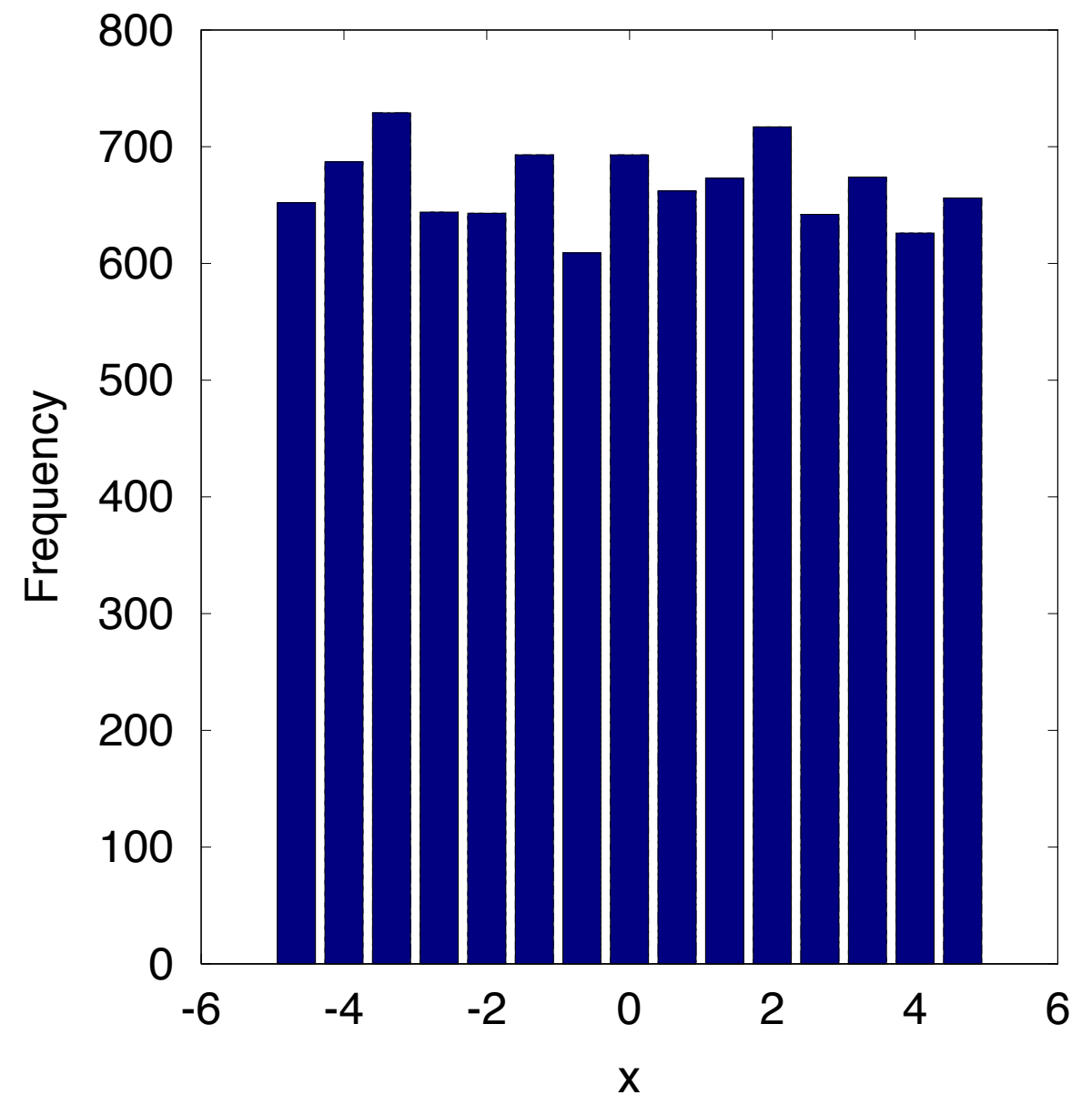
Uniform Distribution

Notation	$X \sim \text{Uniform}(a, b)$
PDF	$f_X(x) = \begin{cases} \frac{1}{b-a} & , a < x < b \\ 0 & , \text{otherwise} \end{cases}$
Mean	$\mu_X = \frac{a+b}{2}$
Variance	$\sigma_X^2 = \frac{(b-a)^2}{12}$

Uniform Distribution



PDF of Uniform(-5,5)

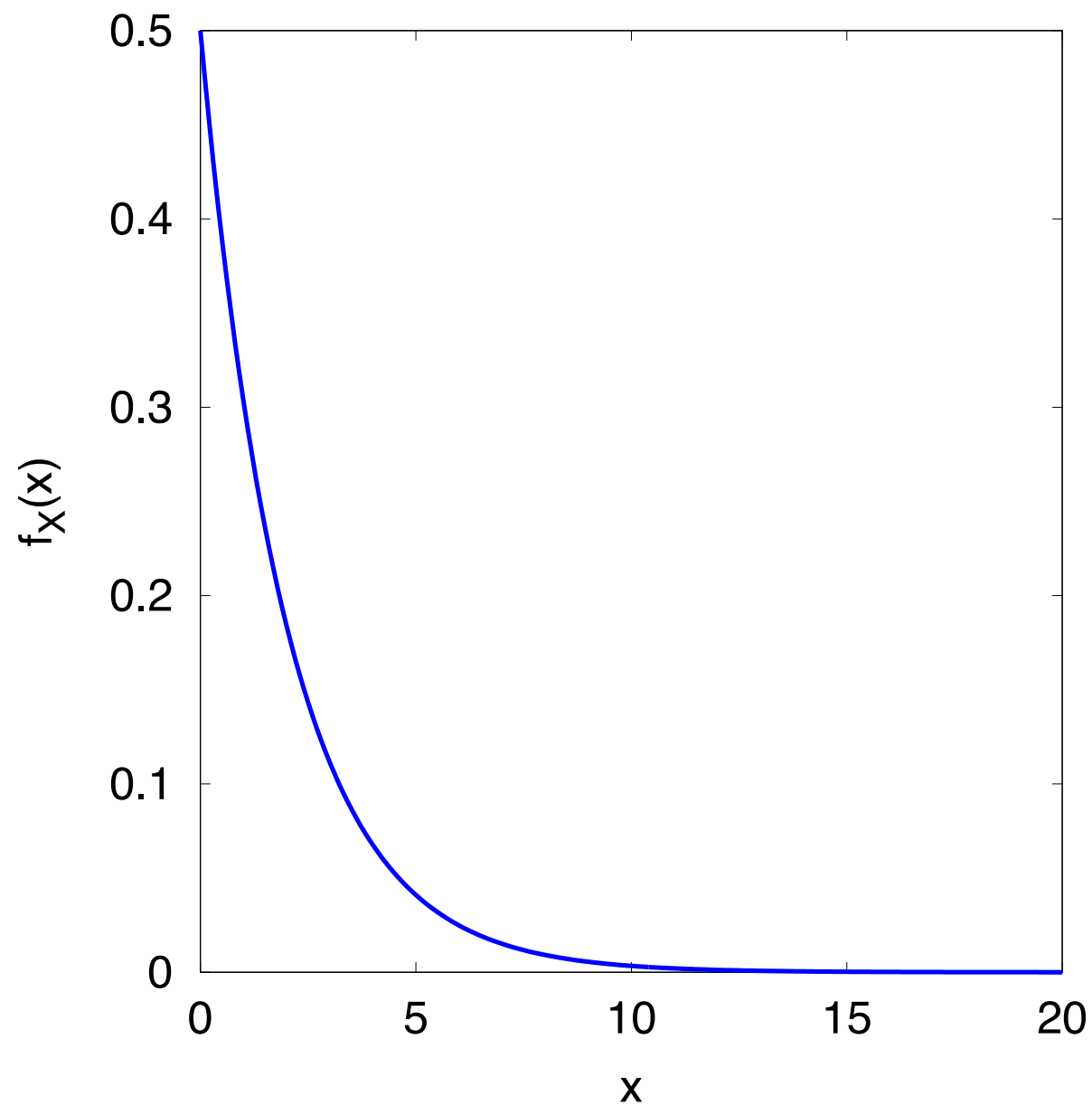


Histogram of samples drawn from Uniform(-5,5)

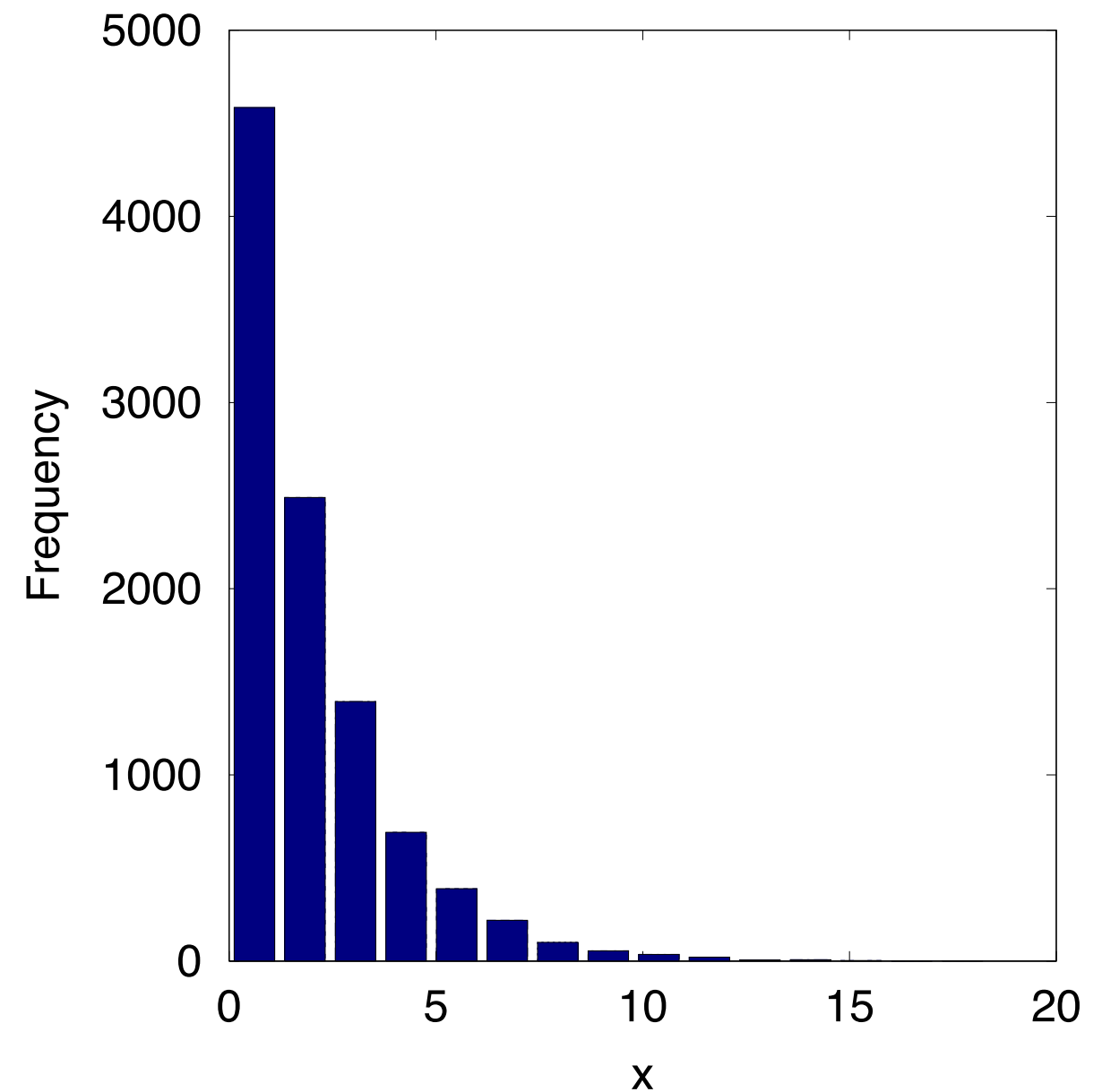
Exponential Distribution

Notation	$X \sim \text{Exp}(\lambda)$
PDF	$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & , x \geq 0 \\ 0 & , \text{otherwise} \end{cases}$
Mean	$\mu_X = \frac{1}{\lambda}$
Variance	$\sigma_X^2 = \frac{1}{\lambda^2}$

Exponential Distribution



PDF of Exp(2)



Histogram of samples drawn from Exp(2)

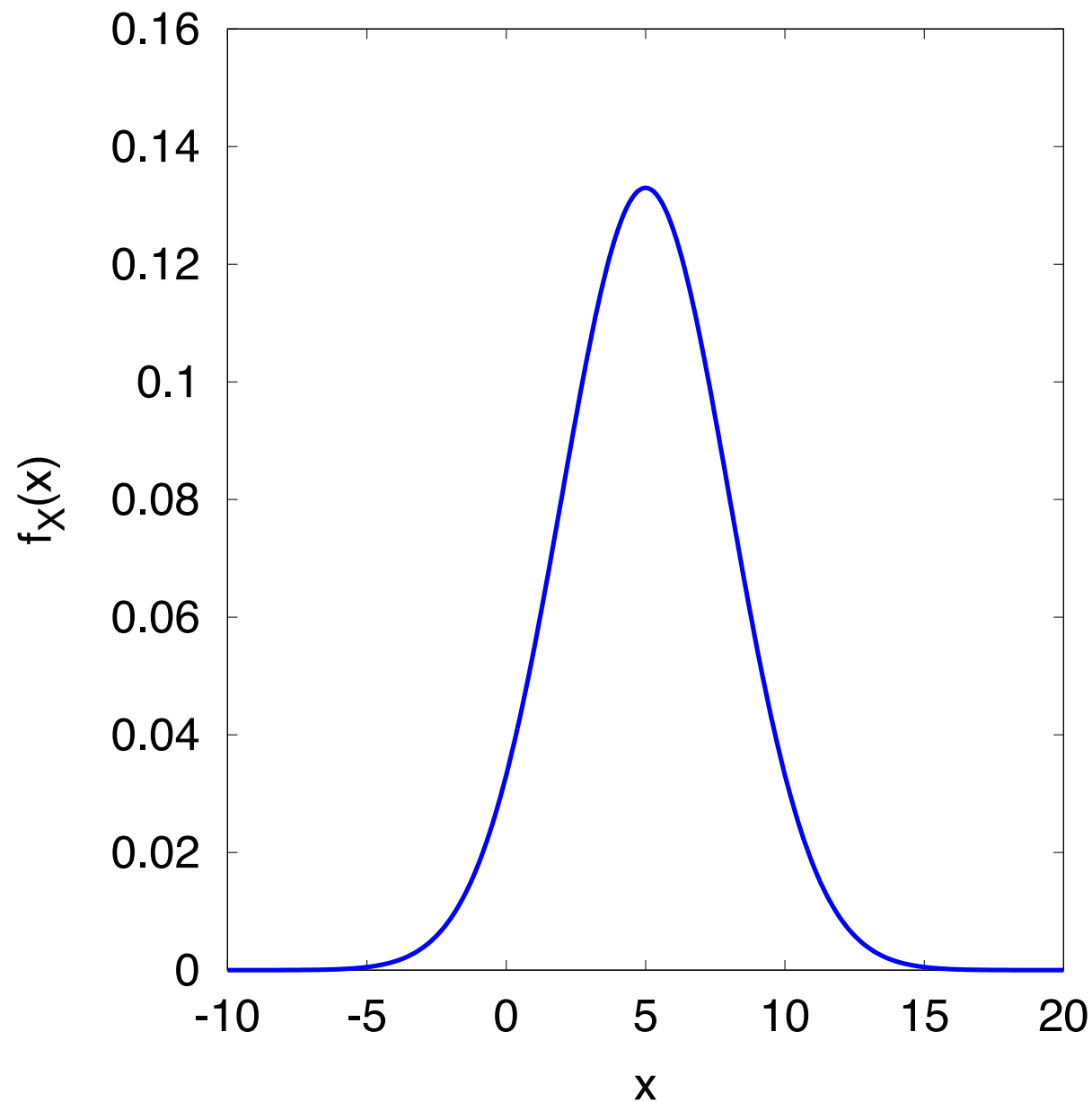
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Normal Distribution

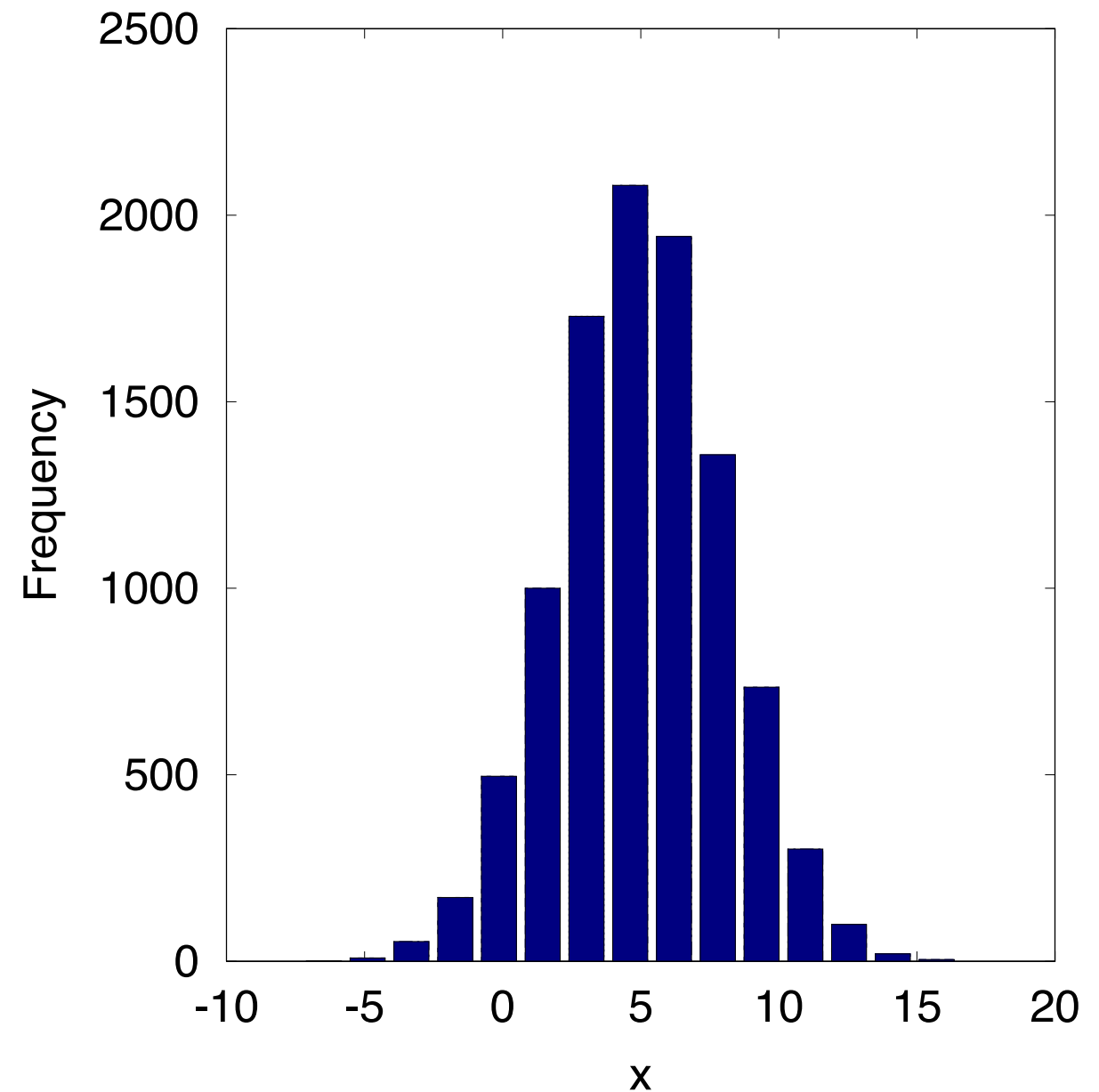
Notation	$X \sim \mathcal{N}(\mu, \sigma^2)$
PDF	$f_X(x) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2} & , -\infty < x < \infty \\ 0 & , \text{otherwise} \end{cases}$
Mean	$\mu_X = \mu$
Variance	$\sigma_X^2 = \sigma^2$

Normal Distribution

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PDF of $N(5,9)$



Histogram of samples drawn from $N(5,9)$

Normal Curve

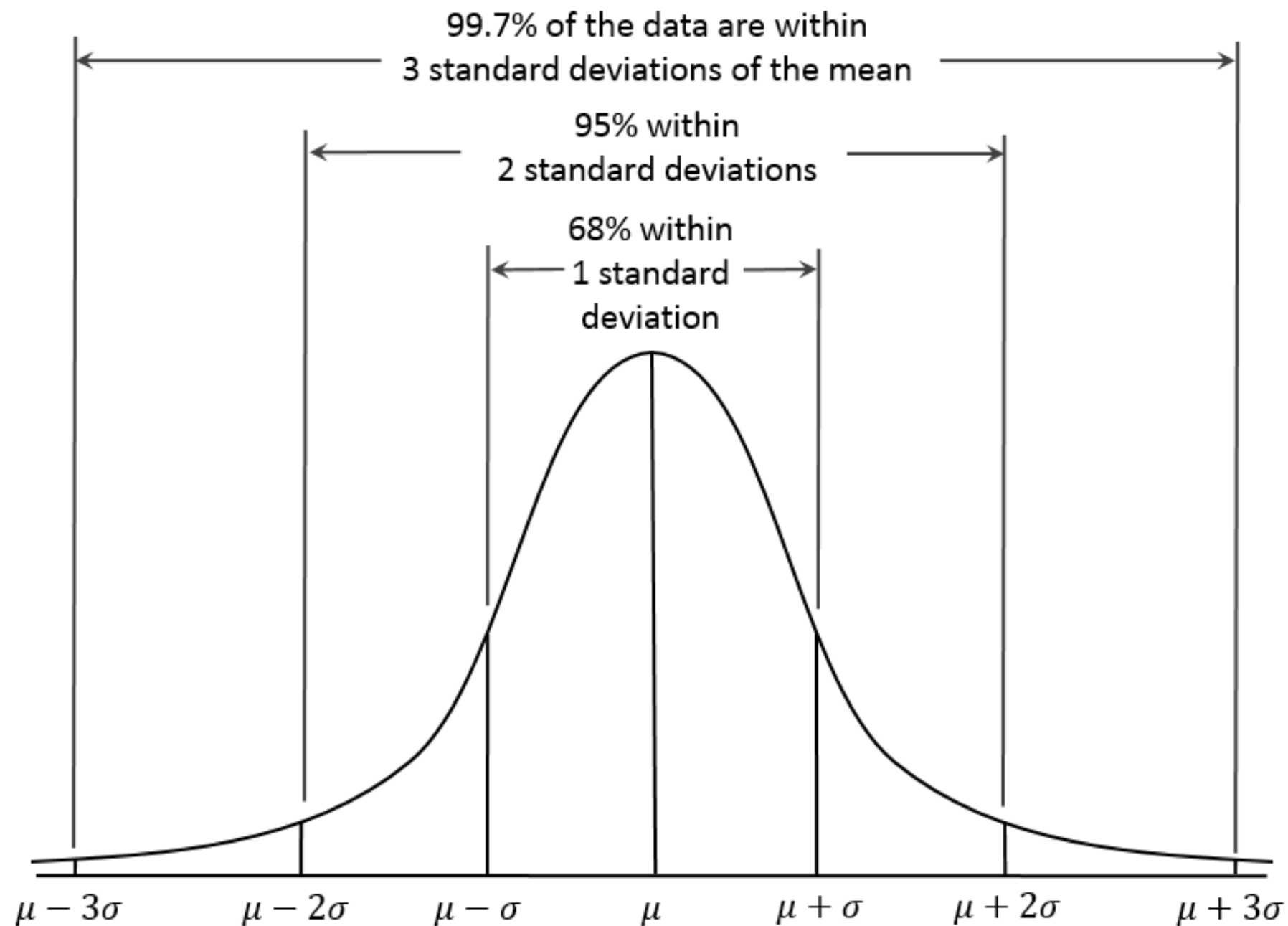


Image Source: Dan Kernler, https://en.wikipedia.org/wiki/File:Empirical_Rule.PNG

Standard Normal

- Normal population with mean 0 and standard deviation 1
- Converting normal population with mean μ and standard deviation σ to a standard normal population

$$z = \frac{x - \mu}{\sigma}$$

- This value is called the “z-score” of x

Standard Normal Table

Standard Normal Probabilities

- For standard normal

$$P(Z \leq z) = \Phi(z)$$

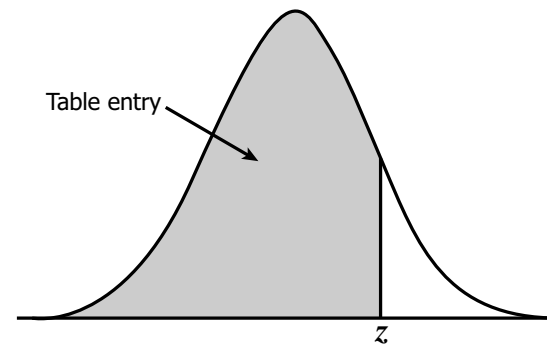


Table entry for z is the area under the standard normal curve to the left of z .

- For $X \sim \text{Normal}(\mu, \sigma^2)$

$$\begin{aligned} P(X \leq x) &= P\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right) \\ &= P\left(Z \leq \frac{x - \mu}{\sigma}\right) \\ &= \Phi\left(\frac{x - \mu}{\sigma}\right) \end{aligned}$$

$\frac{x - \text{mean}}{\text{stan}}$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916

Source:

<http://www.stat.ufl.edu/~athienit/Tables/Ztable.pdf>

Example 1

- Aluminum sheets used to make beverage cans have thickness (in mm) that are normally distributed with mean 10 and standard deviation 1.3. A particular sheet is 10.8 mm thick. Find the z-score

$$z = \frac{x - \mu}{\sigma} = \frac{10.8 - 10}{1.3} = 0.6154 \#$$

- The thickness of a certain sheet has a z-score of -1.7. Find the thickness of the sheet in the original units of mm.

$$-1.7 = \frac{x - 10}{1.3} \quad x = 7.79 \#$$

Example 2

- Find the area under the normal curve to the left of $z = 0.47$

Standard Normal Probabilities

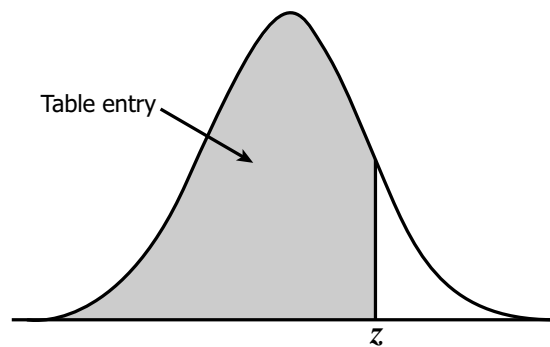


Table entry for z is the area under the standard normal curve to the left of z .

$$P(Z \leq 0.47) = \Phi(0.47) \\ = 0.6808 \#$$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441

Example 2

- Find the area under the normal curve to the left of $z = -0.35$

Standard Normal Probabilities

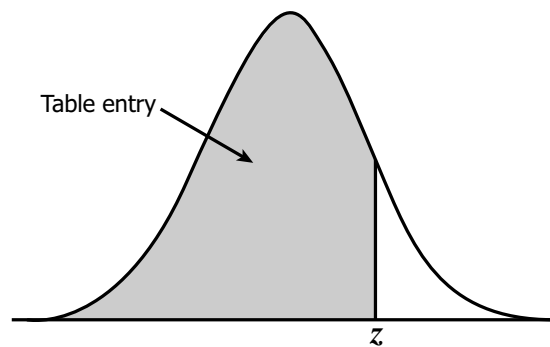


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
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1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441

$$\phi(-z) = 1 - \phi(z)$$

$$\phi(-0.35) = 1 - \phi(0.35)$$

$$\phi(-0.35) = 1 - 0.6368 = 0.3632$$

Example 2

- Find the area under the normal curve to the right of $z = 1.38$

$$P(Z \geq 1.38) = 1 - \Phi(1.38)$$

$$= 1 - 0.9162$$

$$= 0.0838$$

Standard Normal Probabilities

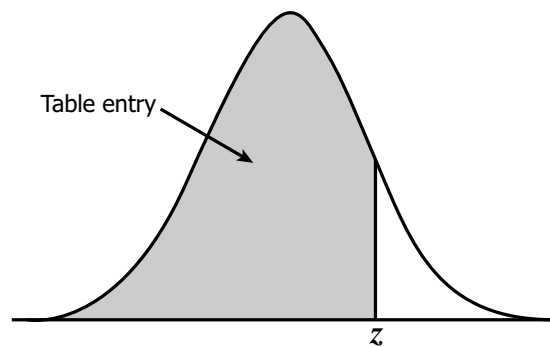


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
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0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
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1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
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1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441

Example 2

- Find the area under the normal curve to between $z = 0.71$ and $z = 1.28$

$$\begin{aligned}
 P(0.71 < Z < 1.28) &= P(Z \leq 1.28) - P(Z \leq 0.71) \\
 &= \Phi(1.28) - \Phi(0.71) \\
 &= 0.8997 - 0.7611 \\
 &= 0.1386 \#
 \end{aligned}$$

Standard Normal Probabilities

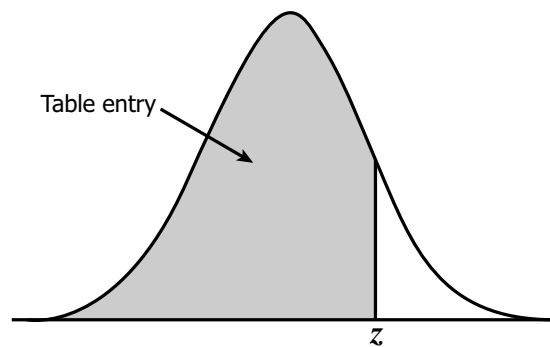
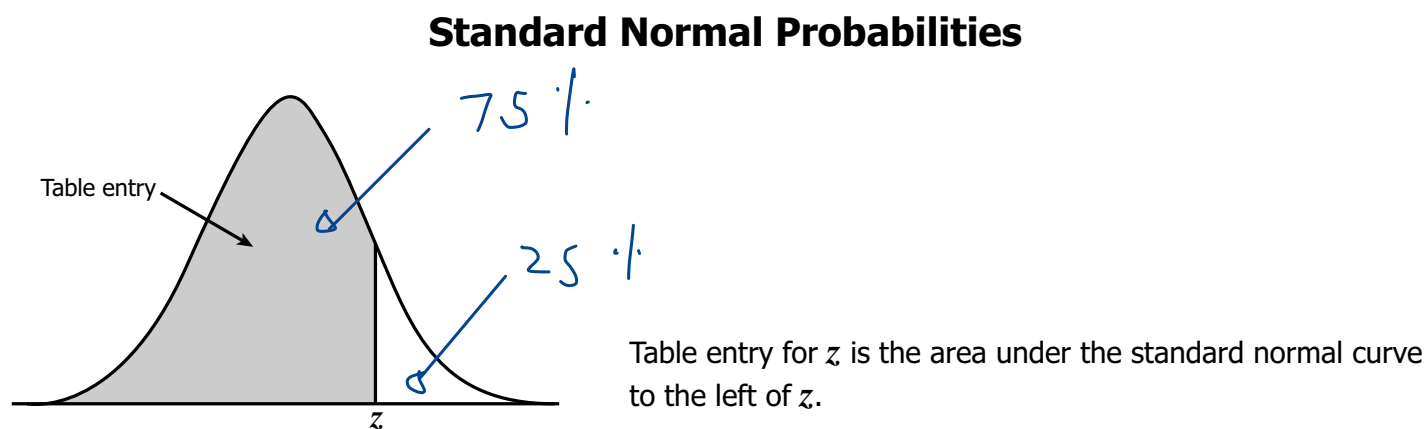


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
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1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
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1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441

Example 3

- What z-score corresponds to the 75th percentile, the 25th percentile, and the median?



75th : $z \approx 0.675$
 25th : $z \approx -0.675$
 50th : $z = 0$
 (median)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
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0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441

Example 4

- Lifetime of batteries in a certain application are normally distributed with mean 50 hours and standard deviation 5 hours. Find the probability that a randomly chosen battery lasts between 42 and 52 hours

$$X \sim N(50, 25)$$

$$P(42 < X < 52) = P\left(\frac{42-50}{5} < Z < \frac{52-50}{5}\right)$$

$$= P(-1.6 < Z < 0.4)$$

$$P(-1.6 < Z < 0.4) = P(Z \leq 0.4) - P(Z \leq -1.6)$$

$$= \Phi(0.4) - \Phi(-1.6)$$

$$= \Phi(0.4) - (1 - \Phi(1.6))$$

$$= 0.6554 - 0.9452$$

$$= 0.6066$$

Standard Normal Probabilities

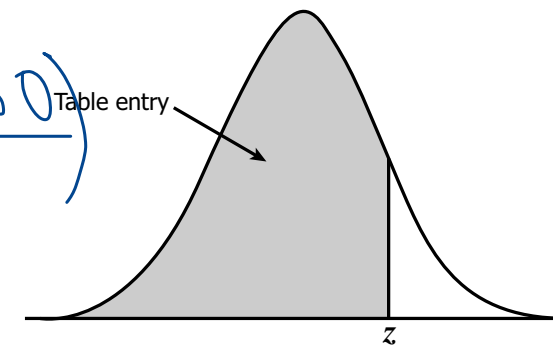


Table entry for z is the area under the curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686

Example 5

- A process manufactures balls whose diameters are normally distributed with mean 2.50 cm and standard deviation 0.008 cm. Specifications call for the diameter to be in the interval 2.5 ± 0.01 cm. What proportion of the balls will meet the specification?

$$X \sim N(\mu, \sigma^2)$$

$$X \sim N(2.50, 0.008^2)$$

$$P(2.49 < X < 2.51)$$

$$= P\left(\frac{2.49 - 2.50}{0.008} < Z < \frac{2.51 - 2.50}{0.008}\right)$$

$$= P(-1.25 < Z < 1.25)$$

$$= \Phi(1.25) - \Phi(-1.25)$$

$$= 0.8944 - (1 - 0.8944) = 0.7888$$

Standard Normal Probabilities

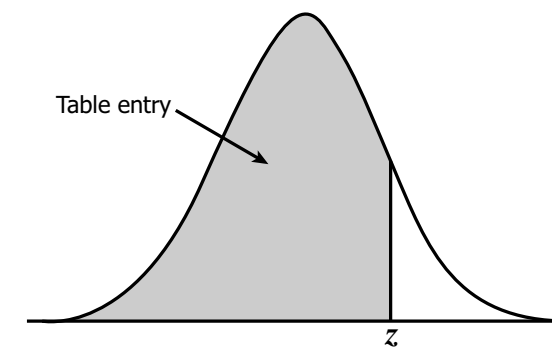


Table entry for z is the area under the curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.527
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.567
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.606
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.644
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.680
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.715
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.748
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.779
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.807
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.834
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.857
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.879
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.898
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.914
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.929
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.941